

Day 3->hacker rank

Task 1

You are given the firstname and lastname of a person on two different lines. Your task is to read them and print the following:

Hello firstname lastname! You just delved into python.

Function Description

Complete the `print_full_name` function in the editor below.

`print_full_name` has the following parameters:

- `string first`: the first name
- `string last`: the last name

Prints

- string: 'Hello *firstname lastname*! You just delved into python' where *firstname* and *lastname* are replaced with *first* and *last*.

Input Format

The first line contains the first name, and the second line contains the last name.

Constraints

The length of the first and last names are each ≤ 10 .

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The first line contains the first name, and the second line contains the last name.

Constraints

The length of the first and last names are each ≤ 10 .

Sample Input 0

Ross

Taylor

Sample Output 0

Hello Ross Taylor! You just delved into python.

Explanation 0

The input read by the program is stored as a string data type. A string is a collection of characters.

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• string last: the last name

Prints

- string: 'Hello *firstname lastname*! You just delved into python' where *firstname* and *lastname* are replaced with *first* and *last*.

Input Format

The first line contains the first name, and the second line contains the last name.

Constraints

The length of the first and last names are each ≤ 10 .

Sample Input 0

```
Ross
Taylor
```

Sample Output 0

```
Hello Ross Taylor! You just delved into python.
```

Explanation 0

The input read by the program is stored as a string data type. A string is a collection of characters.

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```
1 # Complete the 'print_full_name' function below.
2 #
3 # The function is expected to return a STRING.
4 # The function accepts following parameters:
5 # 1. STRING first
6 # 2. STRING last
7
8
9
10 def print_full_name(first, last):
11     # Write your code here
12     print(f"Hello {first} {last}! You just delved into python.")
13
14
15 if __name__ == '__main__':
16     first_name = input()
17     last_name = input()
18     print_full_name(first_name, last_name)
```

Line: 17 Col: 43

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Prints

- string: 'Hello *firstname lastname*! You just delved into python' where *firstname* and *lastname* are replaced with *first* and *last*.

Input Format

The first line contains the first name, and the second line contains the last name.

Constraints

The length of the first and last names are each ≤ 10 .

Sample Input 0

```
Ross
Taylor
```

Sample Output 0

```
Hello Ross Taylor! You just delved into python.
```

Explanation 0

The input read by the program is stored as a string data type. A string is a collection of characters.

You have earned 10.00 points! You are now 15 points away from the 3rd star for your python badge. 63% 95/110

Congratulations

You solved this challenge. Would you like to challenge your friends? [f](#) [t](#) [in](#) [Next Challenge](#)

Test case 0 ✓ Compiler Message

Test case 1 ✓ Success

Input (stdin) [Download](#)

```
1 Ross
2 Taylor
```

Expected Output [Download](#)

```
1 Hello Ross Taylor! You just delved into python.
```

Task 2

Read a given string, change the character at a given index and then print the modified string.

Function Description

Complete the *mutate_string* function in the editor below.

mutate_string has the following parameters:

- string string*: the string to change
- int position*: the index to insert the character at
- string character*: the character to insert

Returns

- *string*: the altered string

Input Format

The first line contains a string, *String*.

The next line contains an integer ,position the index location and a string , separated by a space.

Sample Input

STDIN Function

----- -----

abracadabra s = 'abracadabra'

5 k position = 5, character = 'k'

Sample Output

Abrackdabra

Explanation:

We have seen that lists are mutable (they can be changed), and tuples are immutable (they cannot be changed).

Let's try to understand this with an example.

You are given an immutable string, and you want to make changes to it.

Example

```
>>> string = "abracadabra"
```

You can access an index by:

```
>>> print string[5]
```

a

What if you would like to assign a value?

```
>>> string[5] = 'k'
```

Traceback (most recent call last):

File "<stdin>", line 1, in <module>

TypeError: 'str' object does not support item assignment

How would you approach this?

- One solution is to convert the string to a list and then change the value.

Example

```
>>> string = "abracadabra"
```

```
>>> l = list(string)
```

```
>>> l[5] = 'k'
```

```
>>> string = ''.join(l)
```

```
>>> print string
```

abrackdabra

- Another approach is to slice the string and join it back.

Example

```
>>> string = string[:5] + "k" + string[6:]
```

```
>>> print string
```

abrackdabra

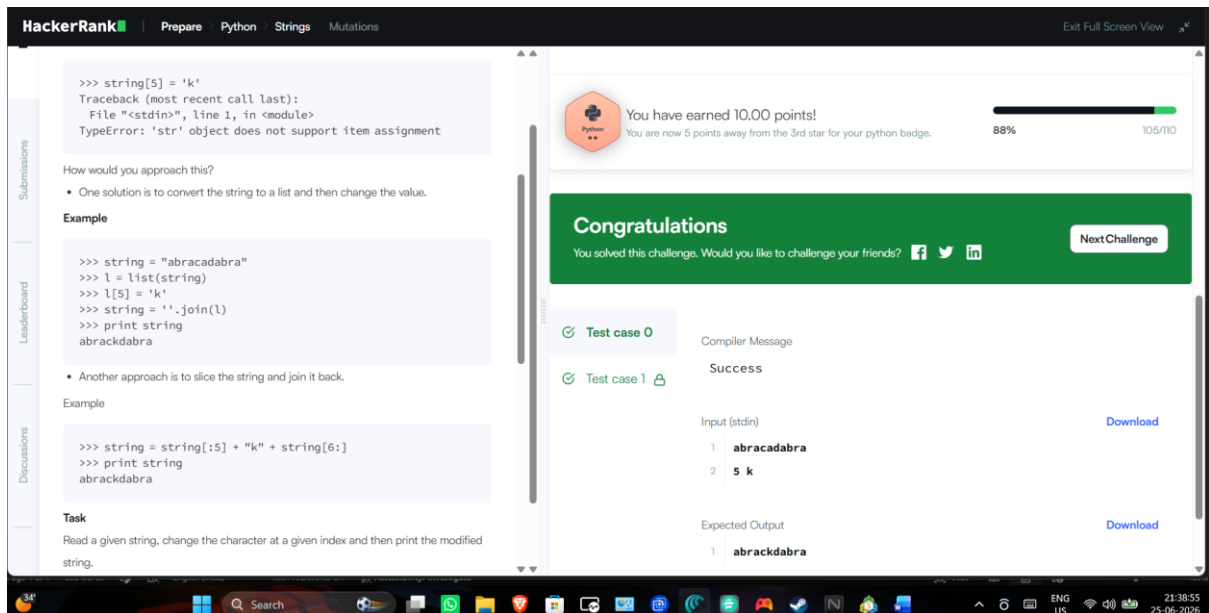
The screenshot shows the HackerRank interface for the 'Mutations' problem. On the left, there's a sidebar with 'Submissions', 'Leaderboard', and 'Discussions' tabs. The main area is divided into two columns. The left column contains a traceback for an error: 'TypeError: 'str' object does not support item assignment'. Below this, it asks 'How would you approach this?' and suggests converting the string to a list. An 'Example' section shows the correct approach using slicing:

```
>>> string = "abracadabra"
>>> l = list(string)
>>> l[5] = 'k'
>>> string = ''.join(l)
>>> print string
abrackdabra
```

. Another approach is also mentioned:

```
>>> string = string[:5] + "k" + string[6:]
>>> print string
abrackdabra
```

. The right column shows a Python solution with a function `mutate_string` and a test script. The function takes a string, position, and character, converts the string to a list, changes the character at the given position, and joins the list back into a string. The test script takes input for the string, position, and character, and prints the result. The bottom of the page has buttons for 'Run Code' and 'Submit Code'.



Task 3

In this challenge, the user enters a string and a substring. You have to print the number of times that the substring occurs in the given string. String traversal will take place from left to right, not from right to left.

NOTE: String letters are case-sensitive.

Input Format

The first line of input contains the original string. The next line contains the substring.

Constraints

$$1 \leq \text{len}(\text{string}) \leq 200$$

Each character in the string is an ascii character.

Output Format

Output the integer number indicating the total number of occurrences of the substring in the original string.

Sample Input

ABCD CDC

CDC

Sample Output

2

Concept

There are a couple of new concepts:

In Python, the length of a string is found by the function `len(s)`, where `s` is the string.

To traverse through the length of a string, use a *for* loop:

for `i` in `range(0, len(s))`:

```
    print (s[i])
```

A range function is used to loop over some length:

`range (0, 5)`

Here, the range loops over `0` to `4`. `5` is excluded.

The screenshot shows the HackerRank interface for the 'Find a string' challenge. On the left, the problem description states: 'In this challenge, the user enters a string and a substring. You have to print the number of times that the substring occurs in the given string. String traversal will take place from left to right, not from right to left. NOTE: String letters are case-sensitive. Input Format: The first line of input contains the original string. The next line contains the substring. Constraints: $1 \leq \text{len}(\text{string}) \leq 200$. Each character in the string is an ascii character. Output Format: Output the integer number indicating the total number of occurrences of the substring in the original string. Sample Input: ABCDCDC, CDC. Sample Output: 2. Concept: There are a couple of new concepts:'. On the right, a Python 3 code editor shows the following solution:

```
def count_substring(string, sub_string):
    count = 0
    for i in range(len(string)-len(sub_string)+1):
        if string[i:i+len(sub_string)] == sub_string:
            count = count+1
    return count

if __name__ == '__main__':
    string = input().strip()
    sub_string = input().strip()
    count = count_substring(string, sub_string)
    print(count)
```

At the bottom right, there are buttons for 'Run Code' and 'Submit Code'.

This screenshot shows the same HackerRank interface after the code has been submitted and executed successfully. At the top right, a message says 'You have earned 10.00 points!' with a progress bar at 5% (1/5/220). A large green banner in the center says 'Congratulations' and 'You solved this challenge. Would you like to challenge your friends?'. Below this, a list of test cases is shown, all marked as passed (green checkmarks):

- Test case 0
- Test case 1
- Test case 2
- Test case 3
- Test case 4
- Test case 5

On the right side, the 'Compiler Message' section shows 'Success'. Below it, the 'Input (stdin)' is displayed as:

```
1 ABCDCDC
2 CDC
```

The 'Expected Output' is shown as:

```
1 2
```

Buttons for 'Download' are provided for both the input and the expected output. At the bottom right, there is a 'Next Challenge' button.